

```

clear, clc

Fs = 1000;
Tpwm = 1/Fs;
Tsim = Tpwm/100;

B = [2.20050890435928e-05,4.40101780871856e-05,2.20050890435928e-05];
A = [1,-1.99630021102880,0.996303145040676];

Be = [0,-8.14529968572852e-08,8.70284222406413e-05];
Ae = [1,-1.99637300531677,0.996375956500999];

Gz = tf(B, A, Tsim)
Gze = tf(Be, Ae, Tsim)

simOut = sim('dados.slx');

figure;
plot(simOut.arx, 'b-', 'LineWidth', 1.5);
hold on;
plot(simOut.esp, 'r--', 'LineWidth', 1.5);
hold off;

xlabel('Tempo (s)');
ylabel('Tensão');
title('Comparação arx/espaco de estados');
legend('ARX', 'espaco de estados');

```

Gz =

$$\frac{2.201e-05 z^2 + 4.401e-05 z + 2.201e-05}{z^2 - 1.996 z + 0.9963}$$

Sample time: 1e-05 seconds

Discrete-time transfer function.

Gze =

$-8.145e-08 z + 8.703e-05$

 $z^2 - 1.996 z + 0.9964$

Sample time: 1e-05 seconds

Discrete-time transfer function.

Warning: Function functions has the same name as a MATLAB built-in. We suggest you rename the function to avoid a potential name conflict.

